

Chapter

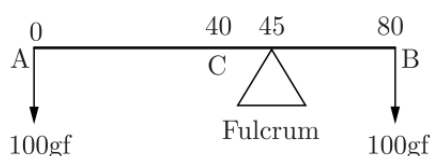
Force : Turning Forces and Uniform Circular Motion

1. A moment of couple has a tendency to rotate the body in an anticlockwise direction. Then the moment of couple is taken as:

- (a) positive
- (b) negative
- (c) maximum
- (d) zero

SQP 2025

2. A metal rod AB of length 80 cm is balanced at 45 cm from the end A with 100 gf weights suspended from the two ends.



(a) If this rod is cut at the centre C, then compare the weight of AC to the weight of BC.

(b) Give a reason for your answer in (a)

SQP 2025

3. A moment of couple has a tendency to rotate the body in an anticlockwise direction. Then the moment of couple is taken as:

- (a) positive
- (b) negative
- (c) maximum
- (d) zero

SQP 2024

4. Calculate the force which will produce a moment of force of 1575 dyne-cm, when the perpendicular distance between point of application of force and turning point is 45 cm.

- (a) 75 dyne
- (b) 45 dyne
- (c) 35 dyne
- (d) 25 dyne

SQP 2024

5. Which of the following is a class III lever?

- | | |
|----------------------|-------------------|
| (a) Pair of scissors | (b) Wheelbarrow |
| (c) Crowbar | (d) Human forearm |

MAIN 2024

6. A door lock is opened by turning the lever (handle) of length 0.2 m. If the moment of force produced is 1 Nm, then the minimum force required is

- | | |
|----------|-----------|
| (a) 5 N | (b) 10 N |
| (c) 20 N | (d) 0.2 N |

MAIN 2024

7. When a stone tied to a string is rotated in a horizontal plane, the tension in the string provides force necessary for circular motion. (Centripetal, centrifugal)

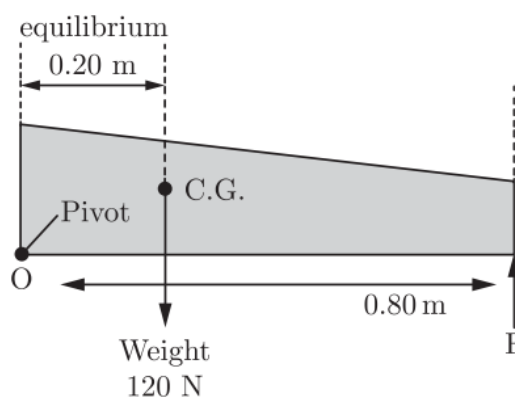
MAIN 2024

8. A hollow ice cream cone has height 6 cm.

- 1. Where is the position of its centre of gravity from the broad base?**
- 2. Will its position change when it is filled completely with honey? Write Yes or No**

MAIN 2024

9. A non uniform beam of weight 120 N pivoted at one end is shown in the diagram below.



Calculate the value of F to keep the beam in equilibrium

MAIN 2024

10. The condition for equilibrium is :

- (a) The resultant of moments of all the forces acting on the body about the turning point should be zero.
- (b) The resultant of all the forces acting on the body be zero only.
- (c) Both (a) and (b)
- (d) None of the above

COMP 2024

11. A handle with a screw jack is used so that the required force decreases. (Long/short)

SQP 2023

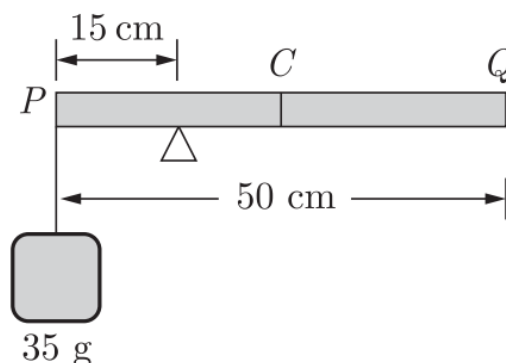
12. What is meant by equilibrium ? State the necessary conditions for a body to be in equilibrium.

SQP 2023

13. A uniform see-saw 4 m long is supported at its centre, Ram weighing 50 kg sits 1 m from the centre of see-saw, where should Geeta weighing 40 kg sit on the other side of the see-saw so as to balance it horizontally?

SQP 2023

14. In the given diagram a uniform half metre scale PQ is balanced horizontally across a knife edge. Calculate (a) the mass of the scale, (b) the force on the knife-edge, (c) state the direction in which the force on the knife-edge, acts.



COMP 2023

15. The centre of gravity of a cricket ball is at:

- (a) at any point on its surface
- (b) at its bottom touching the ground
- (c) its geometric centre
- (d) its top most point

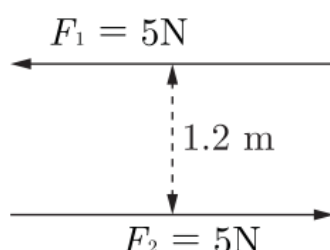
MAIN 2023

16. A couple of 15 N force acts on a rigid body, such that the arm of couple is 85 cm. Calculate the moment of couple in S.I. system.

- (a) 25.00 dyne
- (b) 15.80 dyne
- (c) 37.50 dyne
- (d) 12.75 dyne

COMP 2023

17. Study the diagram given below and calculate the moment of couple.



- (a) 2 Nm
- (b) 1 Nm
- (c) 6 Nm
- (d) 4 Nm

COMP 2023

18. What type of motions are exhibited by a vehicle and its wheels?

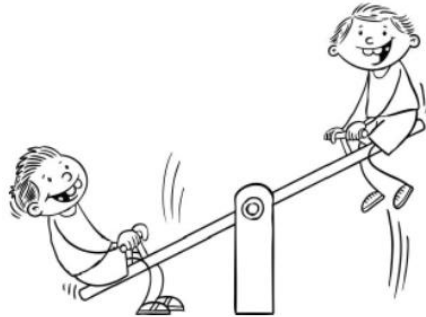
- (a) Translatory and rotatory
- (b) Rotatory
- (c) Translatory
- (d) None of these

SQP 2022

19. A uniform metre scale balances horizontally on a knife-edge placed at 55 cm mark when a mass of 25 g is suspended from one end. Draw a diagram of this arrangement. Calculate the mass of the scale.

MAIN 2022

20. Manoj of mass 55 kg and Ashok of mass 40 kg are sitting on a see saw at a distance of 2 m and 1.5 m respectively from the centre of the see saw in an amusement park as shown in the figure. Is the see saw in rotational equilibrium?



- (a) data incomplete (b) Yes
(c) No (d) None of the above

COMP 2022

21. A 50 cm uniform ruler is freely pivoted at 15 cm mark which balances horizontally when an object of weight 40 dyne is hung from the 2 cm mark.

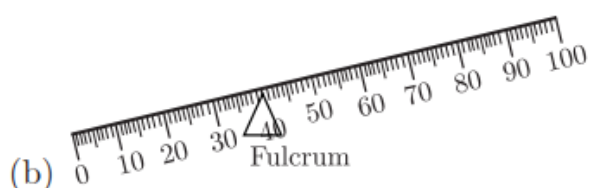
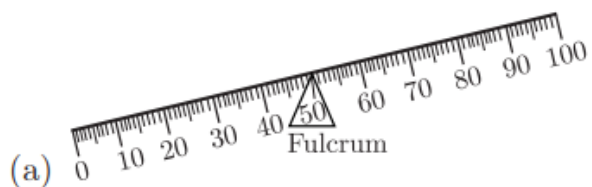
Draw a force diagram of the arrangement and calculate the weight of the ruler. Also calculate the force with which pivot is pressed.

SQP 2021

22. The diagram below shows the balanced position of a meter scale.



Which one of the following diagrams shows the correct position of the scale, when it is supported at the centre?





MAIN 2021 SEM-1

23. The relation between CGS and SI unit of moment of force is

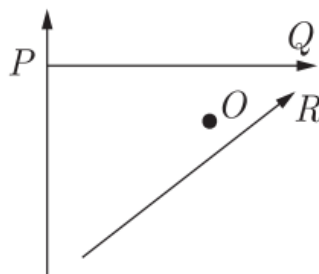
- (a) $1 \text{ Nm} = 10^5 \text{ dyne cm}$
- (b) $1 \text{ Nm} = 10^5 \text{ dyne}$
- (c) $1 \text{ Nm} = 10^7 \text{ dyne cm}$
- (d) $1 \text{ dyne-cm} = 10^7 \text{ Nm}$

SEM-1 2021

24. Can the centre of gravity of a body be outside it? If yes, give one example.

MAIN 2021

25. P , Q , and R are three coplanar forces which act as shown in the figure.



A point O lies in the same plane, $P = Q = R = 6 \text{ N}$.

- (i) Which force has the least moment about O ?
- (ii) Which force has the greatest moment about O ? Give a reason.

COMP 2021

Solution

1.

Ans: (a) Positive

Explanation:

When a couple tends to rotate a body in the **anticlockwise direction**, the **moment of the couple is considered positive**. Conversely, if the rotation is in the **clockwise direction**, the moment is taken as **negative**. Therefore, since the rotation is anticlockwise, the **moment of the couple is positive**.

Thus, the correct option is **(a)**.

2.

Ans:

(a) Weight of AC < weight of BC

(b) The rod balances at 45 cm from A even though equal 100 gf weights are hung at A and B. This shows that the rod's own centre of gravity lies on the side of B, between C and B. Hence more of the rod's weight lies in BC than in AC, so the weight of AC is less than the weight of BC.

3.

Ans: (a) Positive

Explanation:

By sign convention, if a couple tends to rotate a body **anticlockwise**, its moment is taken as **positive**.

If it tends to rotate the body **clockwise**, its moment is taken as **negative**. Here the tendency is anticlockwise, so the moment of the couple is **positive**.

4.

Ans: (c) 35 dyne

Explanation:

The formula for the moment of force is given as:

Moment of force $M = F \times d$

Given:

$$M = 1575 \text{ dyne-cm}$$

$$d = 45 \text{ cm}$$

$$F = \frac{M}{d} = \frac{1575}{45} = 35 \text{ dyne}$$

So, the required force is **35 dyne**.

5.

Ans: (d) Human forearm

Explanation:

In a **class III lever**, the **effort (force)** is applied **between** the **fulcrum** and the **load**.

In the human forearm:

- Fulcrum: **elbow joint**
- Effort: **biceps muscle** pulling on the radius (midway)
- Load: weight in the **hand/palm**

Since effort lies between fulcrum and load, the human forearm is a **class III lever**.

6.

Ans: (a) 5 N

Explanation:

Moment of force (torque) is given by:

$$\text{Moment} = \text{Force} \times \text{Perpendicular distance}$$

Here,

$$\text{Moment} = 1 \text{ N}\cdot\text{m}$$

$$\text{Distance (length of handle)} = 0.2 \text{ m}$$

$$1 = F \times 0.2 \Rightarrow F = \frac{1}{0.2} = 5 \text{ N}$$

So, the minimum force required is **5 N**.

7.

Ans:

Centripetal

Explanation:

In circular motion, the necessary force always acts towards the centre of the circle. Here, the tension in the string pulls the stone towards the centre of the circular path, so it provides the centripetal force.

8.

Ans:

1. The centre of gravity lies on the axis of the cone at a height $\frac{h}{3} = \frac{6}{3} = 2$ cm above the broad base.

2. Yes, its position will change when the cone is completely filled with honey.

9.

Ans: 30 N

Explanation:

The beam is in equilibrium when the clockwise moment equals the anticlockwise moment about the pivot.

The weight of the beam (120 N) acts at its centre of gravity, 0.20 m from the pivot, so its moment is:

$$120 \times 0.20 = 24 \text{ N m (clockwise).}$$

The force F acts at 0.80 m from the pivot, producing an opposite (anticlockwise) moment:

$$F \times 0.80.$$

For equilibrium:

$$120 \times 0.20 = F \times 0.80$$

$$F = (120 \times 0.20) / 0.80 = 30 \text{ N.}$$

Thus, the value of F required to keep the beam in equilibrium is 30 N.

10.

Ans: (c) Both (a) and (b)

Explanation:

For a body to be in equilibrium:

1. The **resultant of all forces** acting on it must be **zero** → no linear (translational) motion.
2. The **resultant moment (torque)** of all forces about any point must also be **zero** → no rotational motion.

Only when **both** these conditions are satisfied is the body truly in equilibrium.

11.

Ans: Long

Explanation:

A screw jack requires less effort when the handle is **long**, because a longer handle increases the moment arm, reducing the force needed to turn the screw. Thus, a **long handle** is used.

12.

Ans:

Equilibrium is the state of a body when several forces act on it but they do not produce any change in its state of rest, uniform motion, or uniform rotation.

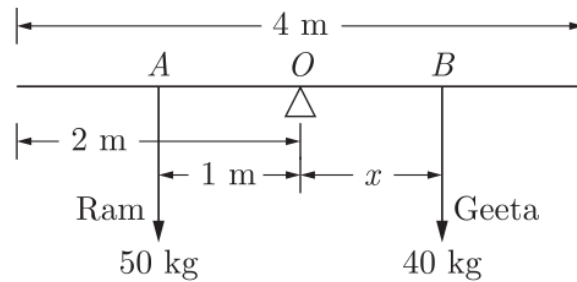
A body is in equilibrium only if the following conditions are met:

- (i) The vector sum of all the forces acting on the body is zero,
i.e., $\sum \mathbf{F}_i = \mathbf{0}$
- (ii) The vector sum of all the torques acting on the body is also zero,
i.e., $\sum \boldsymbol{\tau}_i = \mathbf{0}$

13.

Ans:

Let Geeta sit at a point **B**, at a distance **x** from the centre **O** of the see-saw.



Ram sits **1 m** from the centre.

Using the **principle of moments**:

Anticlockwise moment = Clockwise moment

$$50 \times 1 = 40 \times x$$

$$x = \frac{50 \times 1}{40} = 1.25 \text{ m}$$

So, **Geeta should sit 1.25 m from the centre** to balance the see-saw horizontally.

14.

Ans:

(a) Mass of the scale

Let the mass of the half-metre scale be x g.

Its weight acts at the centre C of PQ.

Taking moments about the knife-edge:

- Clockwise moment = $35 \text{ g} \times 15 \text{ cm}$
- Anticlockwise moment = $x \times 10 \text{ cm}$

For balance,

$$x \times 10 = 35 \times 15 \Rightarrow x = \frac{35 \times 15}{10} = 52.5 \text{ g}$$

So, **mass of the scale = 52.5 g.**

(b) Force on the knife-edge

Total downward force = weight of scale + weight hung at P

$$= 52.5 \text{ gf} + 35 \text{ gf} = 87.5 \text{ gf}$$

So, **force on the knife-edge = 87.5 gf.**

(c) Direction of the force

The knife-edge must support these weights, so the reaction at the knife-edge acts **upwards**, equal to 87.5 gf.

15.

Ans: (c) its geometric centre

Explanation:

A cricket ball is (ideally) a **uniform sphere**, meaning its mass is evenly distributed in all directions. For any uniform sphere, the **centre of gravity** lies at the **geometrical centre**, i.e., the point where any two diameters intersect.

Hence option **(c)** is correct.

16.

Ans: (d) 12.75 dyne

Explanation:

Given:

- Force of couple, $F = 15 \text{ N}$
- Arm of couple $= 85 \text{ cm} = \frac{85}{100} = 0.85 \text{ m}$

Moment of couple:

$$M = F \times \text{arm} = 15 \times 0.85 = 12.75 \text{ Nm}$$

So, the moment of the couple in SI units is **12.75 Nm**.

17.

Ans: (c) 6 Nm

Explanation:

A couple consists of two **equal, opposite, parallel** forces separated by a perpendicular distance (called the **couple arm**).

Here:

Force, $F_1 = F_2 = 5 \text{ N}$

Perpendicular distance between them = 1.2 m

Moment of couple:

$$M = F \times \text{perpendicular distance} = 5 \times 1.2 = 6 \text{ Nm}$$

So, the moment of the couple is **6 Nm**.

18.

Ans: (a) Translatory and rotatory

Explanation:

- The **vehicle as a whole** moves forward along the road → this is **translatory motion**.
- The **wheels** spin about their own axes while the vehicle moves → this is **rotatory (circular) motion**.

So together, a vehicle and its wheels exhibit **both translatory and rotatory motion**.

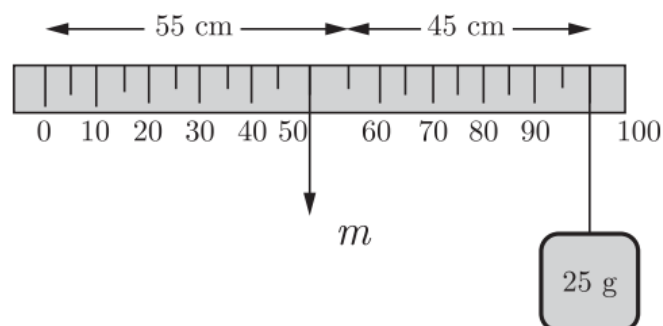
19.

Ans:

Let the mass of the metre scale be m .

Since the scale is uniform, its weight acts at the 50 cm mark.

The knife-edge (pivot) is at the 55 cm mark and a 25 g mass hangs at the 100 cm mark.



Distance of scale's weight from pivot = $55 - 50 = 5 \text{ cm}$

Distance of 25 g mass from pivot = $100 - 55 = 45 \text{ cm}$

Using the principle of moments:

Anticlockwise moment = Clockwise moment

$$m \times 5 = 25 \times 45$$
$$m = \frac{25 \times 45}{5} = 225 \text{ g f}$$

So, the mass of the metre scale is 225 g f.

20.

Ans: (c) No

Explanation:

For rotational equilibrium about the centre (pivot), **sum of clockwise moments = sum of anticlockwise moments.**

Take g as common (so we can compare using mass $\times$ distance):

- Manoj's moment = $55 \times 2 = 110$
- Ashok's moment = $40 \times 1.5 = 60$

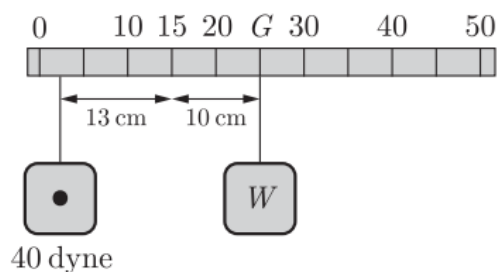
Since $110 \neq 60$, the moments are not balanced.

So, the see-saw is **not** in rotational equilibrium.

21.

Ans:

As per the question we can draw the following figure,



The figure shows the arrangement of forces.

From the principle of moment,

Anticlockwise moment = Clockwise moment

The ruler is uniform, so its weight acts at the 25 cm mark.

Distance of its centre from pivot = $25 - 15 = 10 \text{ cm}$

Distance of 40 dyne weight from pivot = $15 - 2 = 13 \text{ cm}$

$$w \times 10 = 40 \times 13$$

$$w = \frac{40 \times 13}{10} = 52 \text{ dyne}$$

So, **weight of the ruler = 52 dyne**

Force with which the pivot is pressed:

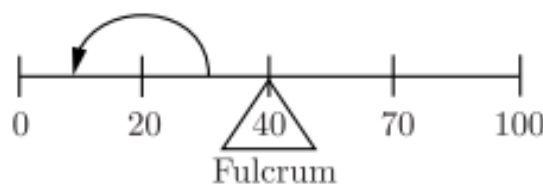
Total downward force = weight of object + weight of ruler

$$= 40 + 52 = 92 \text{ dyne}$$

22.

Ans: (a)

Explanation:



From the first (balanced) position, the fulcrum is nearer the **heavier side**.

Treat the scale in two parts:

- Left part: centre at 20 cm
- Right part: centre at 70 cm

Using moments about the given fulcrum, we get:

$$m \times 20 = m' \times 30 \Rightarrow m = \frac{3}{2}m'$$

So, the **left side is heavier** than the right.

When the scale is now supported at the **centre (50 cm)**, the heavier left side will go **down**, and the lighter right side will go **up**.

Diagram (a) shows exactly this tilt, so option **(a)** is correct.

23.

Ans: (c) $1 \text{ Nm} = 10^7 \text{ dyne}\cdot\text{cm}$

Explanation:

We know:

$1 \text{ newton} = 10^5 \text{ dyne}$

$1 \text{ metre} = 10^2 \text{ cm}$

Therefore,

$1 \text{ N}\cdot\text{m} = 10^5 \times 10^2 = 10^7 \text{ dyne}\cdot\text{cm}$

Thus, **$1 \text{ Nm} = 10^7 \text{ dyne}\cdot\text{cm}$** , and **(c)** is the correct option.

24.

Ans: Yes, the centre of gravity of a body can lie outside the body.

Example:

A uniform ring has its centre of gravity at its geometrical centre — a point that lies outside the ring itself.

25. (i)

Ans: R

Explanation:

Force R is closest to point O , so it produces the **least moment** about O .

25. (ii)

Ans: P

Explanation:

Force P is farthest from point O , so it produces the **greatest moment** about O .